INCH-POUND
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(See 6.9)

DETAIL SPECIFICATION

PLASTIC SHEET, CELL OR CONTINUOUS CAST, ACRYLIC, SHIPBOARD APPLICATION (ILLUMINATION AND SIGNAL LIGHTING)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers acrylic sheet material manufactured by the cell cast, continuous cast or melt (typically referred to as continuously manufactured sheet) process.

NOTE: Acrylic sheet material manufactured by continuous cast meets the requirements of material classified "Melt process category M-1" of ASTM D 4802.

1.2 Classification.

1.2.1 Plastic sheet is of the following types, grades, and thicknesses as specified (see 6.2, tables I and III).

Type RTPS-1 - Red transparent.
Type RTPS-2 - Red transparent, smooth surface finish, one side; patterned pyramidal surface, second side.
Type YTPS-1 - Yellow transparent.

Type GTPS-1 - Green transparent.
Type BTPS-1 - Blue transparent.
Type BTPS-2 - Blue transparent, smooth surface finish, one-side; patterned pyramidal surface, second side.
Type WTLS-1 -
Type WTLS-2 - White translucent (Grades A through G
Type WTLS-3 - (see table III)).
Type WTLS-4 -
Type RTLS-1 - Red translucent.
Type CLEAR-1 - Clear (colorless).
Type CLEAR-2 - Clear (colorless), smooth surface finish, one side; patterned pyramidal surface, second side.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330
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1.2.2 White translucent sheet is of the type and grade specified in table I (see 6.2). The thickness in table I is a requirement of the type and grade specified.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Not applicable.

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 256 - Impact Resistance of Plastics and Electrical Insulating Materials, Standard Test Methods for.
ASTM D 542 - Index of Refraction of Transparent Organic Plastics, Standard Test Method for. (DoD adopted)
ASTM D 570 - Water Absorption of Plastics, Test Methods for. (DoD adopted)
ASTM D 635 - Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position, Test Methods for. (DoD adopted)
ASTM D 648 - Deflection Temperature of Plastics Under Flexural Load, Standard Test Method for. (DoD adopted)
ASTM D 788 - Poly (Methyl Methacrylate) (PMMA) Molding and Extrusion Compounds, Standard Classification System for.
ASTM D 790 - Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials, Standard Test Methods for.
ASTM D 792 - Density and Specific Gravity (Relative Density) of Plastics by Displacement, Standard Test Methods for.
ASTM D 1003 - Haze and Luminous Transmittance of Transparent Plastics, Standard Test Methods for. (DoD adopted)
ASTM D 1925 - Yellowness Index of Plastics, Standard Test Methods for.
ASTM D 2843 - Density of Smoke from the Burning or Decomposition of Plastics, Standard Test Method for. (DoD adopted)
ASTM E 167 - Goniophotometry of Objects and Materials, Standard Recommended Practice for. (DoD adopted)
ASTM E 308 - Computing the Colors of Objects by Using the CIE System, Standard Test Method for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.1 Quality conformance inspection. Quality conformance inspection during production shall consist of the inspection specified in 4.5.

3.2 Materials. The material shall be methacrylate base plastic, manufactured and processed so as to conform to all the requirements of this specification (see 6.2).

3.2.1 Surface finish for all types except for RTPS-2, BTPS-2 and CLEAR-2. Flat sheets shall have a smooth surface finish on both sides.

3.2.2 Surface finish for RTPS-2, BTPS-2 and CLEAR-2. Type RTPS-2, BTPS-2 and CLEAR-2 flat sheets shall have a smooth surface finish on one side and a patterned pyramidal surface on the other side which shall consist of a uniform array of 64 square pyramidal prisms per square inch, each with an included angle of 118 ± 2 degrees at the apex and approximately 0.021 inch high.

3.3 Dimensions.

3.3.1 Length and width. Length and width of sheets shall be as specified in the contract or order (see 6.2).

3.3.2 Thickness. Thickness of flat sheets that are used to fabricate window assemblies of specific shape as specified in the equipment specifications of lighting fixtures shall be as specified in the contract or order to suit contractor's fabrication procedures (see 6.1 and 6.2). Unless otherwise specified in the equipment specifications for lighting fixtures, thickness tolerance of flat sheets which are only cut to specified dimensions without further forming to specific shapes shall be as shown in table I. Thickness of type RTPS-2, BTPS-2 and CLEAR-2 shall be measured from the flat smooth surface to the vertex of the prisms.

TABLE I. Nominal thickness, thickness tolerance and formability diameter.

Nominal thickness (inch) 1/	Thickness tolerances (inch)		Formability diameter (inches) (heated)
	All types and grades except RTPS-2, BTPS-2 and CLEAR-2	RTPS-2, BTPS-2 and CLEAR-2 3/ only	
2/ 0.060	+0.015 -0.019	-	1.5
0.080	+0.014 -0.020	-	2.0
0.100	+0.013 -0.021	-	2.5
0.125	+0.015 -0.025	+0.045 -0.015	3.1
0.150	+0.019 -0.031	-	3.8
0.187	+0.017 -0.033	+0.043 -0.027	4.7
0.220	+0.020 -0.040	-	5.5
0.250	+0.020 -0.040	+0.045 -0.020	6.3

See footnotes at top of next page.

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1/ 0.030, 0.100, 0.150, and 0.220 inch nominal thickness not available for type WTLS-1 through 4.
2/ 0.060 inch nominal thickness not available in types GTPS-1, BTPS-1, and RTPS-1.
3/ Not available in thicknesses 0.060, 0.080, 0.100, 0.150, and 0.220.

TABLE II. Flammability characteristics.

Characteristic	Test specimen (inches) 1/	Applicable test method	Limit
Flammability rate	5 length x 1/2 width x 1/4 thickness	ASTM D 635	1.35 maximum inches/minute
Smoke density rating	1 length x 1 width x 1/4 thickness	ASTM D 2843	50 maximum

1/ A test specimen shall be made to the dimensions shown from the same batch of material offered for delivery.

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TABLE III. Luminous transmittance of white translucent sheet.

Type	Nominal thickness	Luminous transmittance - percent																			
		Grade A			Grade B			Grade C			Grade D			Grade E			Grade F			Grade G		
	Inch	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.	Min.	Nominal	Max.
WTLS-1	0.060	42.0	47.0	52.0	27.0	31.0	35.0	75.0	84.0	93.0	72.0	81.0	90.0	63.0	71.0	79.0	82.0	87.0	92.0	63.0	70.0	77.0
WTLS-2	0.125	29.0	33.0	37.0	14.0	18.0	22.0	66.0	74.0	82.0	58.0	65.0	72.0	49.0	55.0	61.0	76.0	81.0	86.0	63.0	70.0	77.0
WTLS-3	0.187	20.0	24.0	28.0	10.0	13.0	16.0	58.0	65.0	72.0	48.0	54.0	60.0	39.0	44.0	49.0	72.0	77.0	82.0	63.0	70.0	77.0
WTLS-4	0.250	15.0	19.0	23.0	8.0	10.0	12.0	48.0	54.0	60.0	40.0	45.0	50.0	31.0	35.0	39.0	67.0	72.0	77.0	63.0	70.0	77.0

TABLE IV. Optical property values for clear (colorless) sheets.

Property	Value	ASTM test method
Index of refraction	1.49 min., 1.50 max.	ASTM D 542 Refractometric method
Haze	3.0 percent max.	ASTM D 1003
Light transmittance 0.060 - 0.187 in. thick 0.220 - 0.250 in. thick	 91 percent min. 87 percent min.	ASTM E 308
Spectral transmittance at any wave length in the 290 to 330 namometers (nm) band with 0.250 in. thick sheet.	5 percent max. 1/	ASTM D 1003 Using procedure B

1/ When testing other thickness, the value found shall be adjusted to 0.250 inch thickness.

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3.4 Formability. When heated in accordance with the instructions furnished by the manufacturer, the sheets shall be suitable for bending into cylindrical forms of the diameter specified in table I without the appearance of crazing or other surface irregularities.

3.5 Optical uniformity. Plastic sheet shall be free from defects such as imbedded particles, bubbles, scratches, blemishes, and striations, such as will affect the optical properties as specified in 3.5.1 through 3.5.3.

3.5.1 Chips. Maximum allowable size shall be 1/8 inch. Chips approximately 1/8 inch in size shall not have a frequency greater than one chip per 4 square feet of sheet area. Chips less than 1/8 inch shall not form a concentrated pattern that may affect serviceability. Chips out-of-tolerance in size may be knifed off and considered acceptable if the remaining blemish could be removed by polishing.

3.5.2 Embedded particles. Maximum allowable dimension shall be 1/8 inch. Dirt less than 1/32 inch shall be disregarded. The maximum permissible occurrence of dirt with dimensions ranging from 1/32 to 1/8 inch shall be 1 per 4 square feet of sheet area.

3.5.3 Other defects. Minor defects which can be removed by polishing will be acceptable. Defects within 1 inch of the untrimmed edge of the sheet, which do not significantly reduce mechanical strength of the sheet, shall be permitted.

3.6 Specific gravity. The specific gravity of the material shall be 1.18 to 1.20.

3.7 Deflection temperature under load. This deflection temperature under load shall be not less than 185°F.

3.8 Water absorption. The water absorption shall not exceed the limits specified on figure 1 for the appropriate thickness.

3.9 Flammability. Flammability rate and smoke density rating shall not exceed the limits specified in table II.

3.10 Flexural strength. The flexural strength at break shall not be less than 14,000 pounds per square inch (lb/in^2).

3.11 Impact strength (Izod). The Izod impact strength shall be not less than 0.25 foot pounds per inch of notch.

3.12 Optical properties.

3.12.1 Color (chromaticity) and luminous transmittance. The chromaticity in CIE 1/ coordinates and luminous transmittance for the types specified herein shall be as specified in 3.12.1 to 3.12.3, inclusive.

3.12.1.1 Types RTPS-1 and RTPS-2. Red transparent color shall fall within the area bounded by the spectrum locus and y = 0.300, x = 0.985-y as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 6 percent when specimens are tested as specified in 4.6.8. Test samples for RTPS-2 shall be prepared as specified in 4.6.1.3.

1/ Commission Internationale de I'Eclairage (International Commission on Illumination).

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3.12.1.2 Type YTPS-1. Yellow transparent color shall fall within the areas bounded by the spectrum locus and

y = 0.382	x = 0.454 + 0.24Oy
y = 0.790-0.667x	x = 0.698-0.333y

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 50 percent when specimens are tested as specified in 4.6.8.

3.12.1.3 Type GTPS-1. Green transparent color shall fall within the area bounded by the spectrum locus and

y = 0.390-0.171x	x = 0.360-0.080y
	x = 0.650y-0.030

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 20 percent when specimens are tested as specified in 4.6.8.

3.12.1.4 Types BTPS-1 and BTPS-2. Blue transparent color shall fall within the area bounded by the spectrum locus and

y = 0.230	x = 0.133+0.600y
	x = 0.400-y

as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 2 percent when specimens are tested as specified in 4.6.8. Test samples for BTPS-2 shall be prepared as specified in 4.6.1.3.

3.12.1.5 Type WTLS-1 through 4, grades A through G. The optical properties of type WTLS-1 through 4, grades A through G shall be as specified below:

(a) Luminous transmittance. Luminous transmittance shall be as specified in table III.
(b) Yellowness index. Yellowness index shall be 21 maximum.
(c) Diffusion factor. Diffusion factor shall be not less than 0.30.
(d) Filament hiding power. Filament hiding power shall be not less than 0.20.

3.12.1.6 Type RTLS-1. Red translucent color shall fall within the area bounded by the spectrum locus and y = 0.335, x = 0.985-y, as shown on figure 2. Luminous transmittance, using CIE illuminant A, shall be not less than 10 percent when tested as specified in 4.6.8.

3.12.2 Type CLEAR-1. Optical properties for type CLEAR-1 plastic sheets shall be as specified in table IV. The light transmitted by clear plastic sheets shall not be noticeably different in chromaticity from that of the illuminant.

3.12.3 Type CLEAR-2. Optical properties for type CLEAR-2 plastic sheets shall be as specified in table IV except that the test sample shall be prepared as specified in 4.6.1.3.

3.13 Instruction sheet. An instruction sheet shall be furnished with each shipment. Specific information or reference shall be made to the limitations of the material and the necessary precautions to be observed in handling, storing, cutting, drilling, machining, forming, bending, cementing, abrading, polishing, and cleaning.

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4. VERIFICATION

4.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Conformance inspection (see 4.5).

4.3 First article inspection. The sample for first article inspection shall be subjected to the tests of table V.

TABLE V. First article inspection tests.

Tests	Requirement paragraph	Test paragraph
Dimensional examination	3.2, 3.3, 3.5	4.5.1
Formability	3.4	4.6.9
Optical uniformity	3.5	4.5.1
Specific gravity	3.6	4.6.2
Deflection temperature under load	3.7	4.6.3
Water absorption	3.8	4.6.4
Flammability	3.9	4.6.5
Flexural strength	3.10	4.6.6
Impact strength	3.11	4.6.7
Optical properties	3.12	4.6.8

4.4 Sampling for conformance inspection.

4.4.1 Lot. All material of the same type, grade, and thickness manufactured under essentially the same conditions from the same batch of material and offered for delivery at the same time shall be considered a lot for purposes of inspection.

4.4.2 Sampling for visual and dimensional examination. A random sample of sheets shall be selected from each lot offered for examination (see 6.2 and 6.7).

4.4.3 Sampling for tests. A random sample of sheets shall be selected from each lot, except that only one sheet shall be selected from lots of 40 or under. The random sample shall be subjected to the tests specified in 4.6 (see 6.2 and 6.7).

4.5 Conformance inspection.

4.5.1 Visual and dimensional examination. Each sample sheet selected in accordance with 4.4.2 shall be subjected to visual and dimensional examination to determine compliance with 3.2, 3.3 and 3.5. If any sheet in the sample contains one or more defects, as defined in 3.5.1 and 3.5.2, it shall be rejected, and if the number of rejected sheets in the sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample. The thickness shall be determined by the use of a suitable instrument, accurate to 0.001 inch. Five measurements shall be made at random points on the sheet.

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4.5.2 Conformance tests. Conformance tests shall be conducted on the sample sheets selected in accordance with 4.4.3 and shall consist of the tests specified in table VI. Results shall be averaged for each sample sheet to determine conformance with this specification. If the average for any sheet fails to conform with any requirement, this shall be cause for rejection of the lot. Color (chromaticity) and luminous transmittance do not apply to types RTPS-2, BTPS-2, and CLEAR-2.

TABLE VI. Conformance tests.

Test	Requirement paragraph	Test paragraph	Number of determinations
Specific gravity	3.6	4.6.2	1
Deflection temperature under load	3.7	4.6.3	3
Optical properties	3.12	4.6.8	1

4.6 Test procedures.

4.6.1 Test samples.

4.6.1.1 Condition of test samples. Samples shall be tested in the "as received" condition; that is, no special conditioning shall be required unless otherwise specified in the test procedure.

4.6.1.2 Test thickness. Tests shall be made on sample sheets of nominal thickness shown in table I. If sample sheets are tested other than at the nominal thickness, the test results shall be adjusted to the nominal thickness.

4.6.1.3 Test samples for types RTPS-2, BTPS-2, and CLEAR-2. Test samples for RTPS-2, BTPS-2, and CLEAR-2 shall be prepared by either of the following methods:

(a) The patterned surface shall be ground down to the nearest nominal thickness shown in table I and polished to a smooth surface.
(b) The test specimen shall be cast with a smooth surface finish on both sides to the nearest nominal thickness shown in table I from the same batch material as that used for the production sheets under test.

4.6.2 Specific gravity. The specific gravity of the material shall be determined in accordance with ASTM D 792.

4.6.3 Deflection temperature under load. Conditioned test specimens shall be tested. The length of the specimen shall be 5 inches, the depth shall be 0.5 $\pm$ 0.005 inch, and the width shall be the thickness of the original sheet. Thin sheet materials, less than 0.125 inch (namely 0.060, 0.080 and 0.100 inch) shall be arranged to form a composite specimen of convenient thickness. The apparatus shall be similar to ASTM D 648. The conditioned specimen shall be supported as a simple beam on steel supports 4 inches apart with the load applied vertically on the top of the specimen midway between the supports. The 1/2-inch dimension or depth of the beam shall be vertical. The load applied shall be governed by the width of the specimen and shall in each case induce in the specimen a maximum outer fiber stress of 264 lb/in^2. The loads applied for typical specimens are as follows:

Specimen	Load applied midway between the supports
1/4 by 1/2 by 5 inches	2.75 pounds
1/8 by 1/2 by 5 inches	1.38 pounds

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The test specimens shall be in direct contact with a suitable liquid heat transfer medium, liquid petroleum USP, which is agitated during the test. The temperature of liquid shall be raised from normal room temperature at a rate of 3.6 ± 0.9°F per minute. The heat distortion temperature shall be considered the temperature at which the specimen shall deflect 0.010 inch at the center between the supports.

4.6.3.1 Sheets made by the extrusion process of the continuous cast. Extrusion process shall be tested for deflection temperature under load in accordance with the conditioning of test specimens specified for grade 8 material of ASTM D 788.

For 1/8 inch specimens: 185°F
For 1/4 inch specimens: 185°F

The 2 hour period shall be applicable to the sheets of 0.060, 0.080, 0.100 and 0.125 inch thickness. The 4 hour period shall be applicable to the sheets of 0.150, 0.187, 0.220 and 0.250 inch thickness.

4.6.4 Water absorption. The percentage of water absorbed in 24 hours shall be determined in accordance with the water absorption test (for weight and dimensional changes) of ASTM D 570.

4.6.5 Flammability. Tests shall be conducted in accordance with methods shown in table II.

4.6.6 Flexural strength (tested flatwise). The flexural strength shall be determined in accordance with ASTM D 790.

4.6.7 Impact strength. Test for impact strength shall be performed in accordance with ASTM D 256, method A. Specimens shall be tested edgewise.

4.6.8 Tests for optical properties. Tests for optical properties shall be as specified in 4.6.8.1 through 4.6.8.3.

4.6.8.1 Types RTPS-1, RTPS-2, YTPS-1, GTPS-1, BTPS-1, BTPS-2, and RTLS-1. Tests for color and luminous transmittance for types RTPS-1, RTPS-2, YTPS-1, GTPS-1, BTPS-1, BTPS-2, and RTLS-1 shall be conducted in accordance with ASTM E 308. The tests shall be conducted using illuminant A.

4.6.8.2 Type WTLS-1 through 4, Grades A through G. Tests shall be as specified in 4.6.8.2.1 through 4.6.8.2.3 using CIE illuminant C.

4.6.8.2.1 Luminous transmittance. Luminous transmittance tests shall be conducted in accordance with ASTM E 308 using the Hardy-type spectrophotometer. Other apparatus is satisfactory if equivalent results are obtained.

4.6.8.2.2 Yellowness index. Yellowness index tests shall be conducted in accordance with ASTM D 1925.

4.6.8.2.3 Diffusion factor and filament hiding power. Diffusion factor (DF) and filament hiding power (FHP) of a material, when illuminated by parallel light incident normally to the specimen, are defined by:

$$DF = \frac{B_{20} + B_{70}}{2B_5}$$

$$FHP = \frac{B_2}{B_0}$$

Where:

B_0 = Brightness at the normal (straight through) view angle.
B_2 = Brightness at a view angle 2 degrees from normal.
B_5 = Brightness at a view angle 5 degrees from normal.
B_{20} = Brightness at a view angle 20 degrees from normal.
B_{70} = Brightness at a view angle 70 degrees from normal.

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This specimen shall be viewed from the side opposite to that on which the light is incident, and measurements shall be made on a goniophotometer using appropriate instrumentation for recording brightness in accordance with ASTM E 167. Diffusion factor and filament hiding power shall be calculated using the above formulas.

4.6.8.3 Type CLEAR-1 and CLEAR-2. One specimen each shall be tested as specified in table IV.

4.6.9 Formability. The formability of the material shall be determined by heating the material in accordance with the supplier's instructions and by bending it into cylindrical forms of the diameter specified in table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flat plastic sheet covered by this specification is intended for use in fabrication of window assemblies for lighting fixtures for Naval service.

6.1.1 Reference. Reference to this specification on drawings or in product specifications may be made in order to specify the flat material to be used in the fabrication of a part. It should be clearly understood that the requirements apply to the flat material prior to forming or fabrication and not necessarily to the material in its final form.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade and thickness required (see 1.2, tables I and III).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.3).
(d) When a first article sample is required (see 3.1.1).
(e) When evidence of material compliance is required (see 3.2 and 6.6).
(f) Length, width and thickness required (see 3.3).
(g) When a first article test report is required (see 4.3).
(h) Acceptable quality level (see 4.4.2, 4.4.3 and 6.7).
(i) Packaging requirements, if needed (see 5.1).

6.3 Fabricated parts or equipment. Requirements applicable to fabricated parts or equipment incorporating this material should be specified in separate specifications or as part of contract or orders.

6.4 Annealing. All parts fabricated from flat sheets should be heat annealed in accordance with specific instructions supplied by the manufacturer of the flat sheet.

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6.5 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriated for the pending contract.

6.6 Material compliance. Prior to submission of finished items for acceptance, the contractor should submit to the contracting officer or his authorized representative satisfactory evidence that the flat sheet materials, prior to forming into specific shapes of window assemblies as specified in the equipment specification for lighting fixtures for Naval Service, comply with all requirements of this specification. Satisfactory evidence may consist of the submission of test reports of the inspection specified in 3.1 and 3.1.1 or notarized certification to this effect. If certification is submitted, this certification should contain a statement to the effect that test records are available upon request at any time within a time frame of not less than 1 year.

6.7 Lot acceptance and rejection criteria.

6.7.1 Sampling for visual and dimensional examination. The random sample of sheets selected from each lot offered for examination should be in accordance with MIL-STD-105 at inspection level II. The acceptable quality level should be equal to 1.5 percent defective (see 4.4.2).

6.7.2 Sampling for tests. The random sample of sheets selected from each lot should be in accordance with MIL-STD-105 at inspection level S-2, except that only one sheet should be selected from lots of 40 or under. The acceptable quality level should be equal to 1.5 percent defective (see 4.4.3).

6.8 Subject term (key word) listing.

Luminous transmittance
Optical properties
Lighting fixtures
Window assemblies
Methacrylate

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy - SH
(Project 9330-N025)

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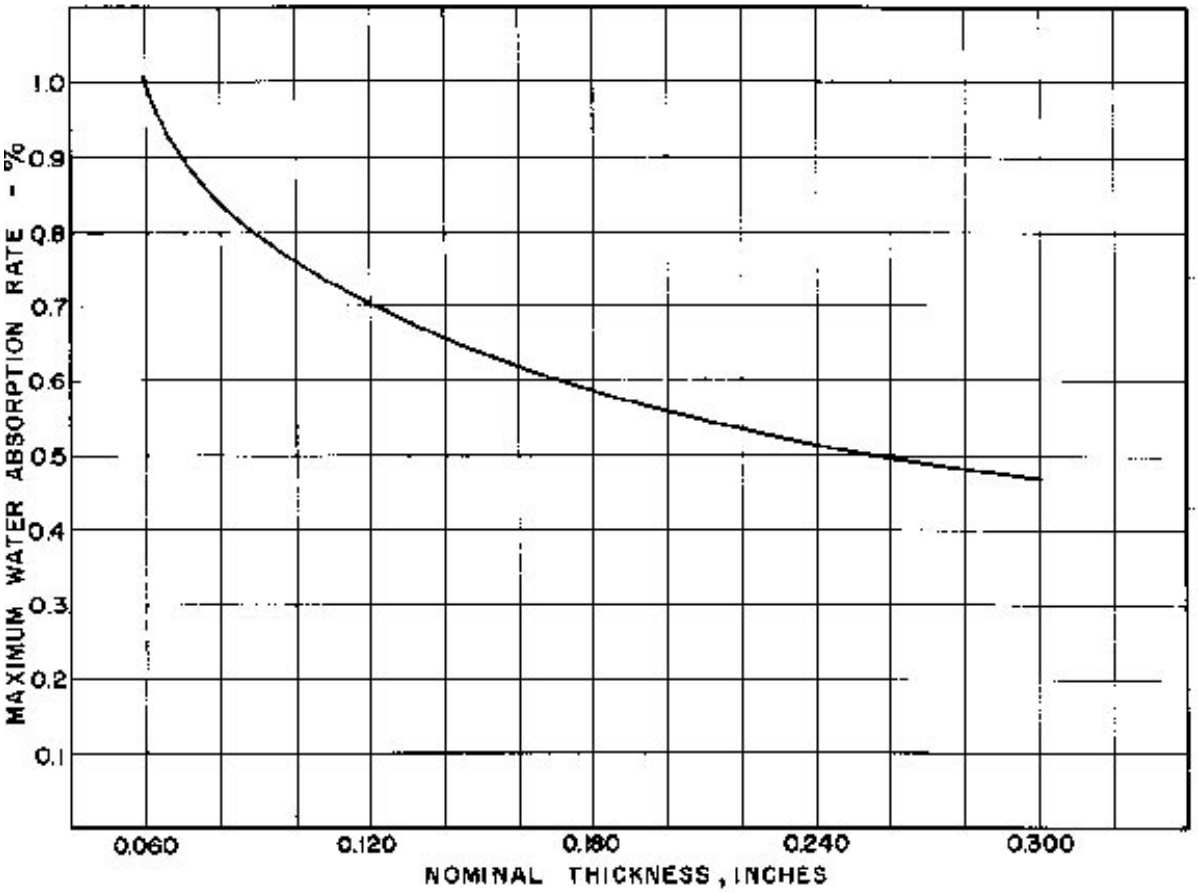


FIGURE 1. Water absorption limits.

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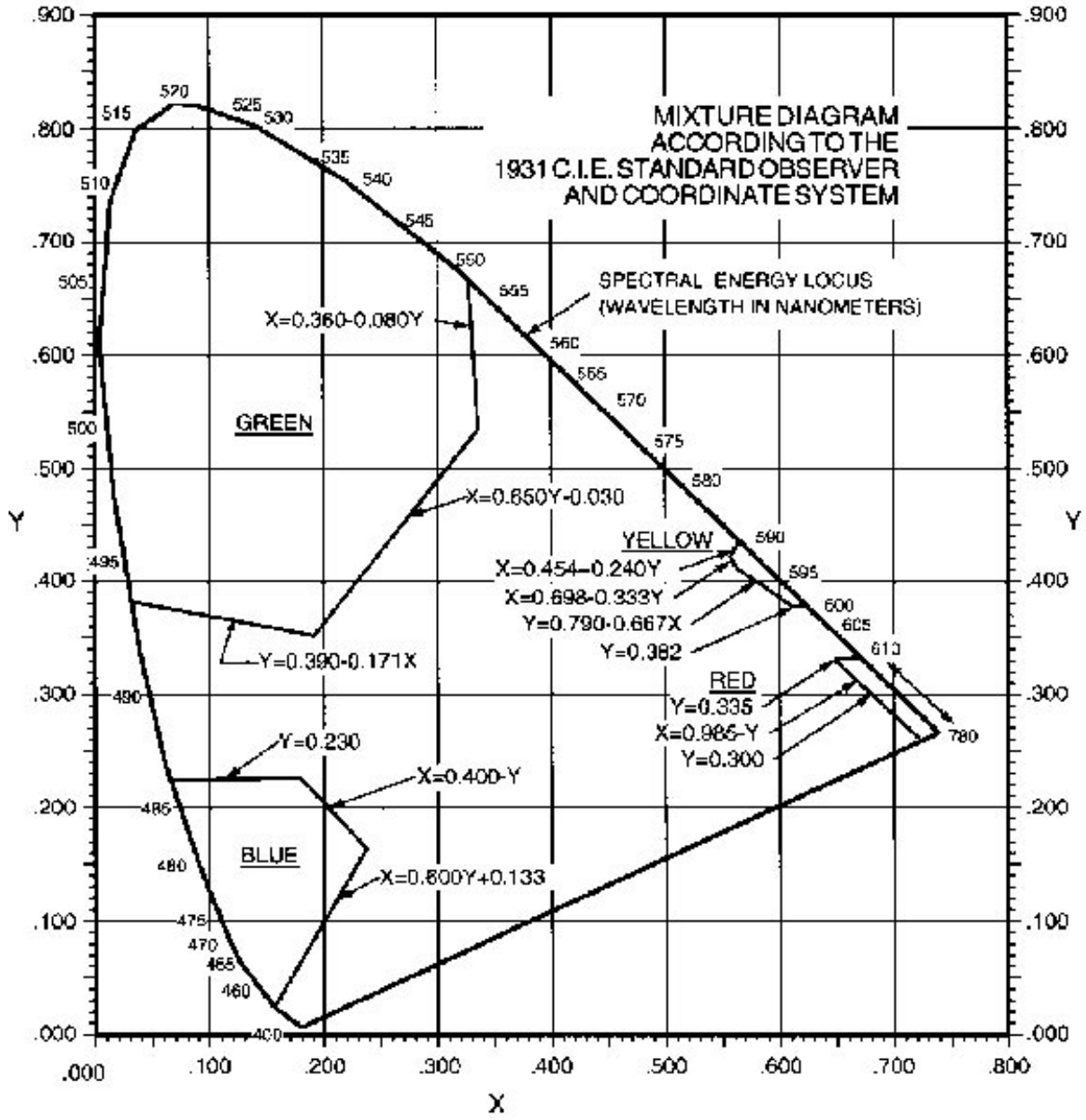


FIGURE 2. Chromaticity diagram illuminate A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-24191D	2. DOCUMENT DATE (YYMMDD) 961122

3. DOCUMENT TITLE
PLASTIC SHEET, CELL OR CONTINUOUS CAST, ACRYLIC, SHIPBOARD APPLICATION (ILLUMINATION AND SIGNAL LIGHTING)

4. NATURE OF CHANGE (identity paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

A. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

A. NAME Technical Point of Contact (TPOC) MR. ALEX VAUGHN, NAVSEA 03E23 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-7440 8-332-7440
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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